Devops(Development + operation)-Methodology

Version Controlling System/SCM – GIT, Bit Backet, Mercurial

But in Market Mostly Used GIT.

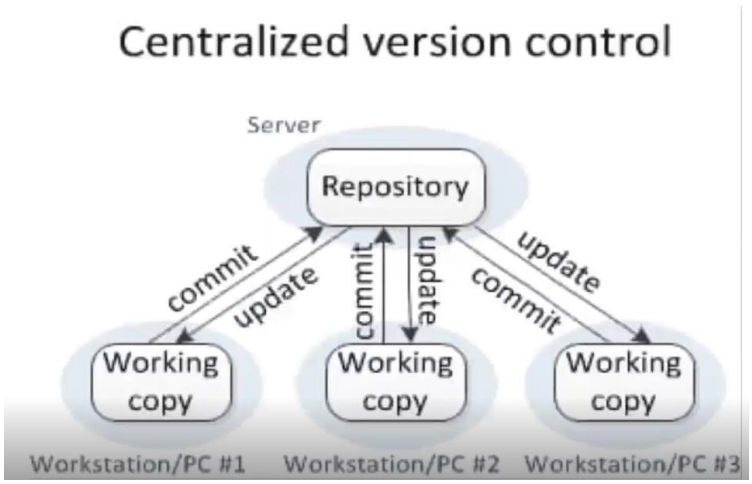
GIT- VCS/source code management (SCM)

GIT is Free. It was Created by Linus Torvalds in 2005 to develop Linux Kernel

GIT Mean Stupid Person. Because during that time It haven't Working Properly.

- **Local Version Control System**
- **CVCS- Centralized Version Control System**
- **Distributed Version Control System**

CVCS- Centralized Version Control System

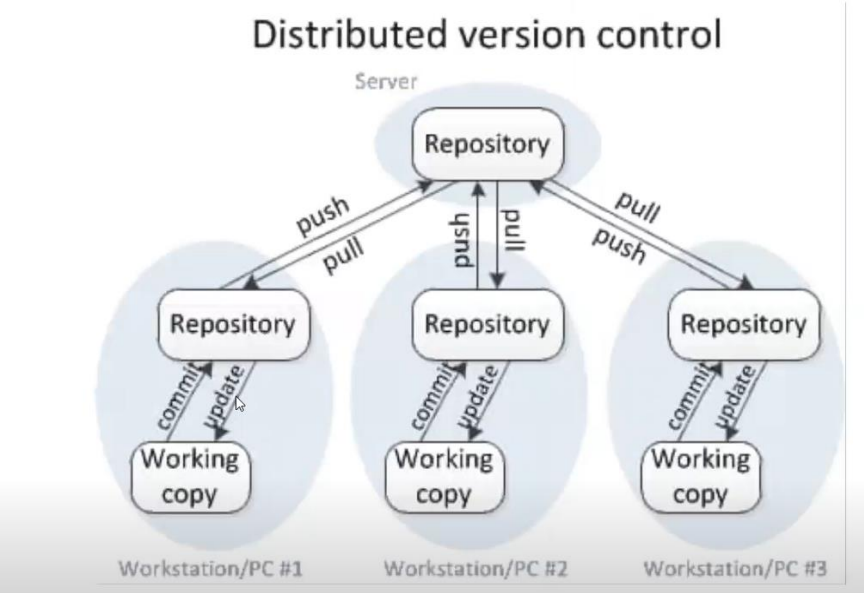


After reaching Repository then application will develop.

Disadvantage of CVCS:

- **Slow Latency**
- **Dependency of Continues Internet**

Distributed Version Control System:



We can commit and update in Local Repository in our local laptop. Here there is no internet dependency.

Internet only using at one time after confirming the file in our local repository.

GIT- LR (Local Repository)

GIThub- RR (Remote Repository)

Install GIT

 # yum install git

 # git –version

 # mkdir git

 # cd git

 # ls -lrt

 # ls -lrta (hidden files)

```
[root@ip-10-0-13-218 git]# ls -lrta
total 0
drwx------ 4 ec2-user ec2-user 85 May 23 10:33 ..
drwxr-xr-x 2 root     root      6 May 23 10:33 .
```

 # git init (initialize the local repository)

 # ls -lrta

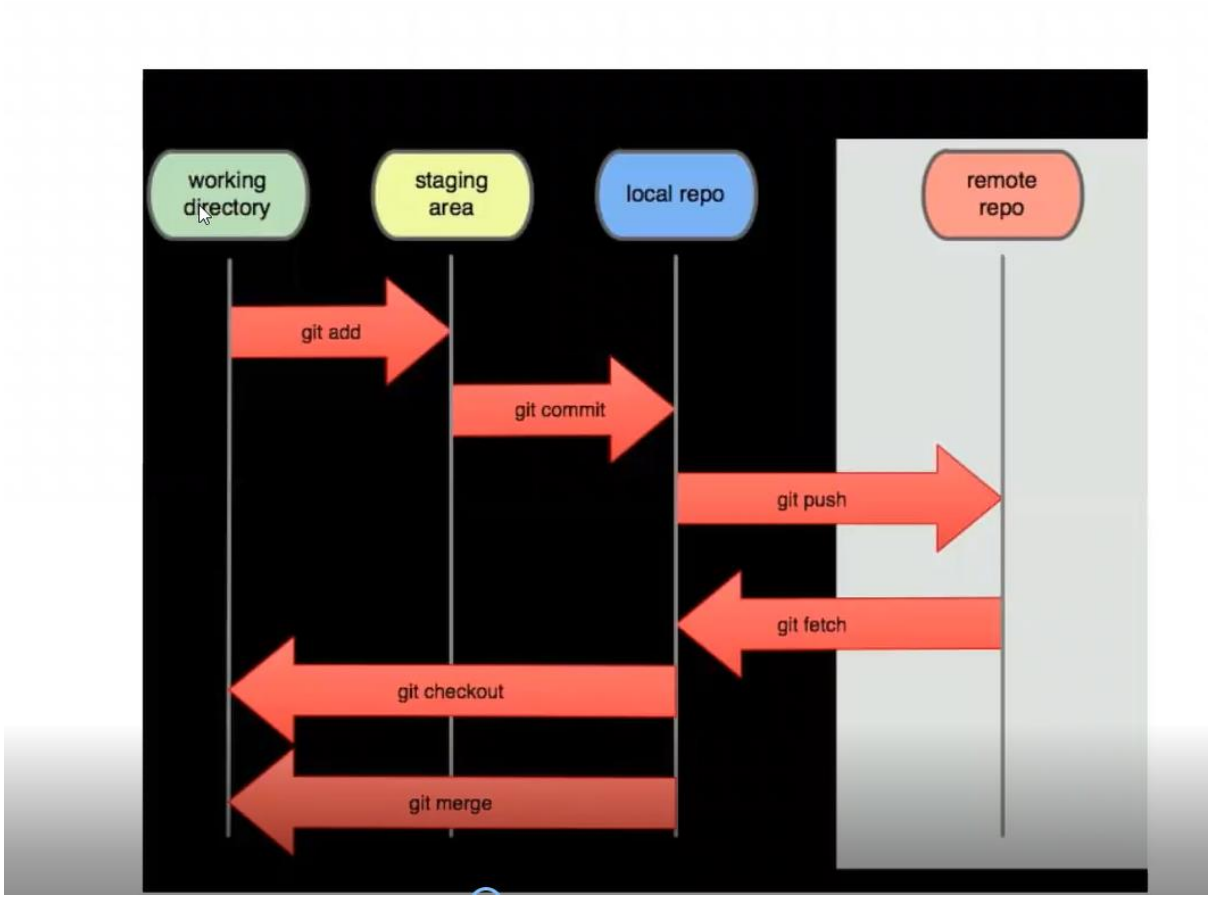
Then it will show .git

#cd .git

 # ls -lrt

/home/ec2-user/git/.git – this git local Repository. (LR)

/home/ec2-user/git – this is working directory (WD)



Working Directory to staging area (we can rollback)

Vi text.txt ----> first -----> wq

ls -lrt

cat vi test

now this file is there in working directory

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

git status -s

```
[root@ip-10-0-13-218 git]# git status -s
?? test.txt
```

git status

```
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        test.txt
```

Now it is in untraced file.

<u>How to Move Staging area:</u>

 # git add . or git add <filename>

 # git status -s

 # git status

 # git config --global user.name "arunkeerthi"

 # git config –global user.email arunsivan31@gmail.com

 # git log

It is showing some error which is not commit to master.

 # git commit -m "description"

 # git log

Now it showing which has committed with proper details.

 # git status

Once we commit everything the status will show "nothing to commit, working tree clear"

Then we do second Commit for the same text file.

 # vi text.txt ------> second ------> wq

#git status -s

(It shows red color M) . (Why it will be shown like we modify same file so that)

#git add .

#git status -s

(It shows Green Color M)

git commit -m “second commit”

#git status -s

git log

(now it will be shown two logs)

git log --oneline (for short)

Git Hub-------

Create repository in GitHub.

Name should be created unique in your GitHub.

#git push

(now it showing some error commit URL)

#git remote add origin URL (name origin (customized name) we give variable for that URL (git hub repro URL))

git remote add origin

git branch

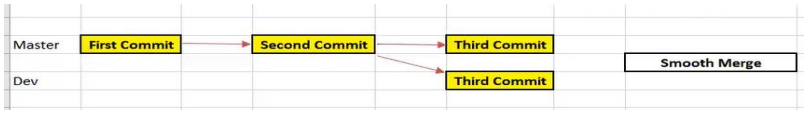
git push origin master (master - branch name)

1. Now it requires username and password
2. We need to give the credential
3. If there asking token, we need to go git hub
4. GitHub→setting→developer setting→personal access tokens→note (customized name) and give it full access→generate token
5. And we take the token number and past it in the password which has come in GitHub.
6. Then go and check the git hub that file will show in git hub repository.

#git log
it will be shown push details too.

BRANCHES

SCENARIO-1 SMOOTH MERGE



- Master it is nothing but original copy
- But real scenario we will not commit in master branch
- Real time mostly we use the feature branch only
- If it everything done by "Maker and Checker", then only we will merge to the master branch.
- Feature branch nothing but we may mean "backup"
- Master Branch only access senior person or admin person

#git branch (to view the branch details)

#git branch dev (To create the new feature Branch, dev-customized name)
#git branch
#git checkout dev (switch to the other branch)
#git branch
#git log
Now we are dev branch
#vi text.txt→third→wq
#git commit -am "third Commit"
#git log or git log --oneline
Now it is showing third commit in dev

Now we go to master branch and check the log what will there?

#git branch
#git checkout master
#git branch

(Then here It will show two commits only)

Now we are going to do merge feature to master branch

Before merging we need to check in which branch are we there. Because we need to do merge at "master branch"

#git branch (check which branch are we there?)

#git merge dev

And view the file we able to see which done in Feature branch

#cat test.txt

#git log

The above-mentioned steps are smooth merging.

```
[root@ip-10-0-13-218 git]# git log
commit 9068f3cf979787d82d2a474e62ef5f91cdc3b774 (HEAD -> master, dev)
Author: arunkeerthi <arunsivan31@gmail.com>
Date:   Mon May 23 14:19:46 2022 +0000

    third commite

commit f24f67b9e5eea8352b4a73cfc17a1bf5421d761e (origin/master, chekout)
Author: arunkeerthi <arunsivan31@gmail.com>
Date:   Mon May 23 13:00:41 2022 +0000

    second commit

commit e6f7e8e1687e7ef31adc623050fa536653ac1038
Author: arunkeerthi <arunsivan31@gmail.com>
Date:   Mon May 23 12:49:37 2022 +0000

    first commit
[root@ip-10-0-13-218 git]#
```

New push the central repository
#git push origin master

SCENARIO-2 CONFLICT MERGE:

 # mkdir merge

 # cd merge

 # ls -lrta

 # git init

#vi merge.txt→ first→wq

Note: untracked file we can't directly commit first time **(for this try it (git commit -am "first commit"))-error will occur**

First time we need to add the file working directory → staging area

#git add .

#git commit -m "first commit"

Then do second commit

#vi merge.txt→ second → wq

#git commit -am "second commit"

#git log

Now Create Branch

#git branch dev

#git log (check the log status)---->here we stand in master only

#git branch

#git checkout branch

#git log -------> (check log it shows both dev and master are same status)

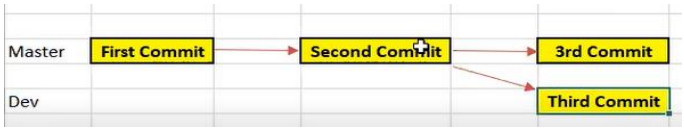
Then do third commit

#vi merge.txt→ third → wq

#git commit -am "third commit"

#git log ------> (here log status show different)

Now some other developer come, and he has directly accessed the master and do some changes.



#git checkout master

#vi merge.txt→3rd line →wq

#git commit -am "3rd commit"

#git log

Now senior developer come and check the details.

#git branch

#git merge dev

It is showing conflict error

```
[root@ip-10-0-13-218 merge]# git branch
  dev
* master
[root@ip-10-0-13-218 merge]# git merge dev
Auto-merging merge.txt
CONFLICT (content): Merge conflict in merge.txt
Automatic merge failed; fix conflicts and then commit the result.
[root@ip-10-0-13-218 merge]#
```

#git log (and check what happen) ---(it showing 3rd commit)that's why conflict happen

 # cat dev (file status)

So finally, conflict happen between the master and feature branch

So that we need to solve the conflict problem

Two options is there to resolve the issue----->Example
1.we can modify the test file but real time we can't do because large numbers of coding will be there)
2.Install merge conflict to arrest the issue

For this case we need to install the merge conflict tool in git.

#git config --global merge.tool vimdiff (vimdiff name)

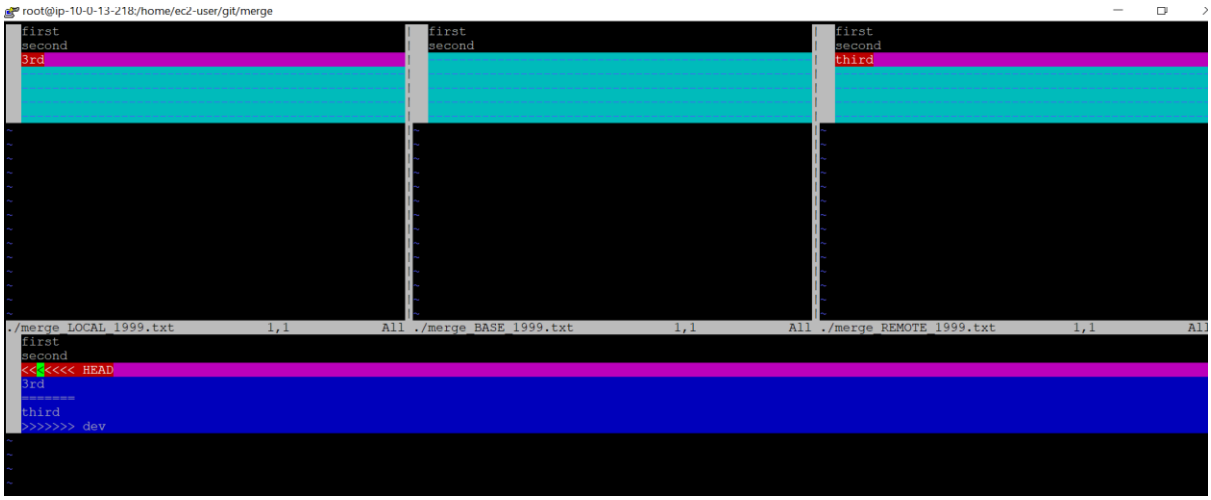
#git config --global merge.conflictstyle diff3 (diff3 style it show three differentiation)

#git config --global mergetool.prompt false (prompting falser to avoid yes or no question)

#git mergetool

(Here merge tool will do auto analysis for both branches)

It is showing like below mentioned tool



Ctrl+ww the Crouser will move then we may edit which ever we want.

Here it is showing the three differentiations

./merge_LOCAL_1999.txt →master, ./merge_BASE_1999.txt→base
./merge_REMOTE_1999.txt→branch

By using this below commands we may confirm.

:diffg LO enter then :wqa

:diffg RE enter then :wqa

:diffg BA enter button then :wqa

ctrl dd

:wqa!

runs from master : git merge feature_branch

choose any commands and resolve the merge conflict

before given the command we need to press esc button around 10times

 # :diff RE enter then :wqa! (example)

#cat dev.txt (Now it showing which we changed in merge conflict)

ls -lrt

one more file also will come that is org file.

```
arun.txt  arun.txt.orig
[root@ip-172-31-33-208 conflict]# ls -lrt
total 8
-rw-r--r-- 1 root root 72 May 24 14:12 arun.txt.orig
-rw-r--r-- 1 root root 19 May 24 14:16 arun.txt
```

#cat arun.txt.org

Here It is showing previous status which we did wrong

#git status (it shows that two file status)

Example:--

```
[root@ip-172-31-15-99 merge]# git status -s
M  test
?? test.orig
[root@ip-172-31-15-99 merge]#
```

#git commit -m "final"

#git status -s

If we want to delete that additional, we may delete

 # rm -Rf <file name>

#git log –oneline ---> (Note: we stand in master only)

Now it is showing five commit details.

When we do Merge Conflict, we will be getting additional Commit.

```
[root@ip-172-31-33-208 conflict]# git log --oneline
ebe6fdf (HEAD -> master) final
aec700f 3rd commit
4f776ac (fb1) third commit
f313276 second commit
b36ed7e first commit
[root@ip-172-31-33-208 conflict]#
```

How to solve the Merge Conflict issue by using Rebase Command:

#git checkout master

#git log

```
[root@ip-172-31-33-208 conflict]# git log
ebe6fdf (HEAD -> master) final
aec700f 3rd commit
4f776ac (fb1) third commit
f313276 second commit
b36ed7e first commit
```

#git checkout fb1

#git log

```
[root@ip-172-31-33-208 conflict]# git log --oneline
4f776ac (HEAD -> fb1) third commit
f313276 second commit
b36ed7e first commit
[root@ip-172-31-33-208 conflict]#
```

Above Mentioned two branch logs it shows commits results are different

So that we need do some changes and equalize both branches commit then only

We will do further in feature branch

By using rebase command we will rectify the issue

Rebase command we should use in feature branch.

 # git rebase master

```
[root@ip-172-31-33-208 conflict]# git rebase master
Successfully rebased and updated refs/heads/fb1.
[root@ip-172-31-33-208 conflict]#
```

 # git log

Then all commit will same as both branches

CHERRY

We can pick the particular commit by using Cheery Command from master to feature branch

 # Mkdir cherry

 # cd cherry

 # git init

#vi text → 1 → wq

 # git add .

 # git commit -m “first commit”

#git log

 # git branch (it show master branch)

 # git branch dev (now Create the feature Branch)

 # git log –oneline (Now commit will show in master as same as feature branch)

 # vi cherry→ 2 → wq

#git commit -m “second commit”

#vi cherry→3→ wq
#git commit -m “third commit”

#git log –oneline ----> (In Master will show three commit)

#git checkout dev

#git log ---> (here showing only one commit)

Then here will show one commit

#git cherry-pick <commit Id> this command we should do in feature branch.

Note : commite id will be change once we pick the master to feature branch

#git log --oneline

```
[root@ip-172-31-33-208 cherry]# git log --oneli
571570e (HEAD -> dev) second commit
ae40d9c first commit
[root@ip-172-31-33-208 cherry]#
```

STASH

Roll back from staging area (buffer location) to working directory (re-apply is possible).

#mkdir stash

#vi stash.txt

#git add .

#git commit -m "first"

 # git log –oneline

#vi stash .txt → just edit :wq

#git stash

#git stash list (then we get the statsh id)

Then again and go to edit some

#vi stash.txt→edit→wq

Then again go stash

#git stash

#git stash list

Here we will two stash details because we have applied two stash.

Move → pop

Copy → Apply

#git stash pop stashID

(now it will move to work directory)

#git stash list

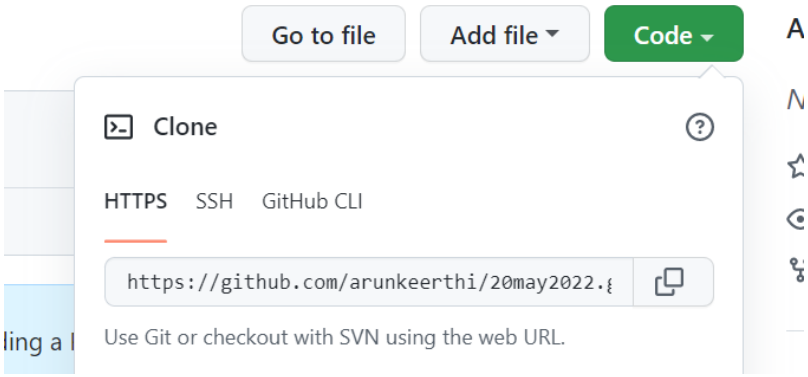
(Now we can check in stash list , id which we use in pop that will be delete in stash list)

Clone:

Backup and recovery - we take the backup for our EC2 server.

Go to git hub and choose any of the project details in repository.

Code→copy url



Copy url and go to git wd

#git clone url

#ls

That will fill show in your working directory.

Pull:

Latest Source Code always available in Github repository

So that the Purpose of enhance or update we pull the source Code from RR to LR

Pull = Fetch - (LR) + Merge - (WD) for the process technically we called us Check Out

(Push = LR + RR ----> Check in)

1.Go to working directory

#git pull url

```
From https://github.com/arunkeerthi/may23
 * branch            HEAD       -> FETCH_HEAD
hint: Pulling without specifying how to reconcile divergent branches is
hint: discouraged. You can squelch this message by running one of the following
hint: commands sometime before your next pull:
hint:
hint:   git config pull.rebase false  # merge (the default strategy)
hint:   git config pull.rebase true   # rebase
hint:   git config pull.ff only       # fast-forward only
hint:
hint: You can replace "git config" with "git config --global" to set a default
hint: preference for all repositories. You can also pass --rebase, --no-rebase,
hint: or --ff-only on the command line to override the configured default per
hint: invocation.
fatal: refusing to merge unrelated histories
```

If showing like this error above mentioned picture because already we have file

So that We need to pull forcefully

#git pull --allow-unrelated-histories url

#esc → : → wq!

#ls -lrt

FORK

Fork is used to transfer the details from git hub one account to another account.

By using git hub repository URL link to see the fork details one account to another account

git push -f url master

#git branch -d dev / force branch delete # git branch -D dev

#git diff (diff between wd to stage area)

#git diff - -staged (stage area to local repo)

#git show (last commit details)

#git show <commit id> (particular commit details)

#git blame <file name> (we know about update of file command details)

#git branch -a (hidden branch)

#git push origin - -delete dev (remote branch delete)

#git rm - -cached file1

 # git restore --staged <file>

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